Math 110 — Week 8

Generalized eigenvectors and Jordan form (Notes Ch. 14–15, LADR 8A–8C, HW8)

§1 Facts used from earlier weeks

V is finite-dimensional over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$), $n = \dim V$. Shorthand: "S kills v" means $Sv = 0$, and "q kills T" means $q(T) = 0$.

(T1) **Rank–nullity.** For linear $T\colon V \to W$: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

Why.

1. Extend a basis $u_1, \dots, u_k$ of $\operatorname{null} T$ to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V. It suffices to show $Tv_1, \dots, Tv_r$ is a basis of $\operatorname{range} T$.
2. Spanning: T of any vector is a combination of the Tu_i (which are 0) and the Tv_j.
3. Independent: if $\sum c_j Tv_j = 0$, then $\sum c_j v_j \in \operatorname{null} T$ is a combination of the u_i, and independence of the basis of V gives all $c_j = 0$.

(T2) **For operators, injective $\iff$ surjective $\iff$ invertible.**

Why. $\operatorname{null} T = 0 \iff \dim \operatorname{range} T = n$ (T1) $\iff \operatorname{range} T = V$ (T3).

(T3) **Dimension shortcuts.**

(a) An independent list of n vectors is a basis.

(b) A subspace of dimension n is all of V.

(c) If $U \cap W = \{0\}$ and $\dim U + \dim W = n$, then $V = U \oplus W$.

(d) $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Why.

1. (a), (b): an independent list extends to a basis, and every basis has length n.
2. (c): a basis of U followed by a basis of W is independent (a relation gives $u = -w \in U \cap W = \{0\}$) and has length n, so it is a basis by (a).
3. (d): extend a basis $x_1, \dots, x_k$ of $U \cap W$ to a basis x_i, u_j of U and, separately, to a basis x_i, w_l of W. Together the x_i, u_j, w_l span $U + W$; it remains to show they are independent.
4. If $x + u + w = 0$ (x, u, w combinations of the x_i, u_j, w_l), then $w = -x - u \in U \cap W$, so w is a combination of the x_i. The basis x_i, w_l of W is independent, so $w = 0$. Then $x + u = 0$, and independence of x_i, u_j gives $x = u = 0$.

(T4) **Products of operators.** If S, T are injective, so is ST. So if a product sends some $v \neq 0$ to 0, some factor is not injective.

Why. $STv = 0 \Rightarrow Tv = 0 \Rightarrow v = 0$.

(T5) **Polynomials in T commute.** $p(T)q(T) = q(T)p(T)$. Hence $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are T-invariant.

Why. Both products equal $(pq)(T)$. If $p(T)v = 0$, then $p(T)(Tv) = T\,p(T)v = 0$. And $T\big(p(T)u\big) = p(T)(Tu) \in \operatorname{range} p(T)$.

(T6) **Fundamental theorem of algebra.** A nonconstant $p \in \mathcal{P}(\mathbb{C})$ factors as $c(z-\lambda_1)\cdots(z-\lambda_m)$ (cited). Over any field, $p(\lambda) = 0 \iff (z-\lambda) \mid p$.

Why. Divide p by $z-\lambda$: the remainder is a constant, and evaluating at λ shows it equals $p(\lambda)$.

(T7) **Eigenvalues.**

(a) λ is an eigenvalue $\iff$ $T - \lambda I$ is not injective.

(b) If $Tv = \lambda v$, then $p(T)v = p(\lambda)v$.

(c) Eigenvectors for distinct eigenvalues are independent (Prop. 54).

(d) Over $\mathbb{C}$ with $V \neq 0$, every operator has an eigenvalue.

Why.

1. (a) is the definition: $Tv = \lambda v$ with $v \neq 0$ means $(T - \lambda I)v = 0$ with $v \neq 0$.
2. (b): $T^k v = \lambda^k v$ by induction; add up the terms of p.
3. (d): take $v \neq 0$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so $p(T)v = 0$ for some $p \neq 0$; p is not constant since $v \neq 0$.
4. Factor $p = c(z-\lambda_1)\cdots(z-\lambda_m)$ (T6). The product $c(T-\lambda_1 I)\cdots(T-\lambda_m I)$ sends $v \neq 0$ to 0, so some $T - \lambda_j I$ is not injective (T4).

(T8) **Upper-triangular matrices.** $\mathcal{M}(T)$ is upper-triangular in $v_1, \ldots, v_n \iff$ every $\operatorname{span}(v_1, \ldots, v_j)$ is invariant. Then the eigenvalues of T are the diagonal entries. Over $\mathbb{C}$ every operator has an upper-triangular matrix.

Why.

1. Column j holds the coordinates of Tv_j, so "upper-triangular" means $Tv_j \in \operatorname{span}(v_1, \ldots, v_j)$ for every j.
2. If $a_{jj} = \lambda$: $T - \lambda I$ maps $\operatorname{span}(v_1, \ldots, v_j)$ into $\operatorname{span}(v_1, \ldots, v_{j-1})$, a smaller space, so it is not injective (T1). So λ is an eigenvalue.
3. If no a_{jj} equals λ: $\mathcal{M}(T - \lambda I)$ is triangular with nonzero diagonal. Its columns are independent: in a relation, the right-most column used is the only one with a nonzero entry in its own diagonal row. So $T - \lambda I$ is injective.
4. Existence over $\mathbb{C}$, by induction on n: take an eigenvalue λ (T7(d)). $U = \operatorname{range}(T - \lambda I)$ is invariant (T5) and $\dim U < n$, since $T - \lambda I$ is not injective (T1).
5. By induction U has a basis $u_1, \ldots, u_m$ with every $\operatorname{span}(u_1, \ldots, u_j)$ invariant. Extend it to a basis $u_1, \ldots, u_m, v_1, \ldots, v_k$ of V.
6. $Tv_j = (T - \lambda I)v_j + \lambda v_j \in U + \operatorname{span}(v_j) \subseteq \operatorname{span}(u_1, \ldots, u_m, v_1, \ldots, v_j)$. So every span of an initial part of the list is invariant.

(T9) **Block diagonal matrices.** If $V = U \oplus W$ with both T-invariant, then in a basis of U followed by one of W, $\mathcal{M}(T) = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$, and $\operatorname{null} T = \operatorname{null} T|_U \oplus \operatorname{null} T|_W$.

Why. $Tu \in U$ is a combination of the U-basis only, so its column is 0 in the W-rows (same for W). If $T(u+w) = 0$, then $Tu = -Tw \in U \cap W = \{0\}$.

(T10) **Change of basis.** If column j of S holds the B-coordinates of the j-th vector of B', then $\mathcal{M}_{B'}(T) = S^{-1}\mathcal{M}_B(T)S$. So similar matrices are exactly the matrices of one operator in two bases.

Why. S converts B'-coordinates to B-coordinates: apply S, then $\mathcal{M}_B(T)$, then convert back with S^{-1}. Every invertible S arises this way, since its columns form a basis.

(T11) **Minimal polynomial (Week 7).** m_T is the monic polynomial of least degree with $m_T(T) = 0$. (a) $q(T) = 0 \iff m_T \mid q$ (Prop. 65). (b) The roots of m_T in $\mathbb{F}$ are exactly the eigenvalues (Prop. 66). (c) $\deg m_T \le n$ (Prop. 67). (d) If U is invariant, $m_{T|_U} \mid m_T$. (e) If $V = U_1 \oplus \cdots \oplus U_r$ with every U_i invariant, m_T is the least common multiple of the $m_{T|_{U_i}}$.

Why. (a) Divide q by m_T; the remainder is 0 at T and has smaller degree, so it is 0. (d) $m_T(T|_U)u = m_T(T)u = 0$, then (a). (e) $q(T) = 0$ exactly when $q(T|_{U_i}) = 0$ for all i, i.e. when every $m_{T|_{U_i}}$ divides q. (b), (c) are proved in the Week 7 handout.

§2 This week

§2.1 Generalized eigenvectors (Ch. 14)

The question. We want the simplest matrix for every operator. With an eigenbasis it is diagonal, but often there are too few eigenvectors.

- $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has eigenvectors only on $\operatorname{span}(e_1)$.
- But $Ne_2 = e_1 \neq 0$ and $N^2 e_2 = 0$: e_2 is sent to 0 by N^2 but not by N.
- Idea of the chapter: allow $(T - \lambda I)^k v = 0$ for some $k \ge 1$ instead of only $k = 1$. Over $\mathbb{C}$ that gives enough vectors for a basis.

Definition (Def. 37 / LADR 8.8, 8.19)**.** $v \neq 0$ is a **generalized eigenvector** for λ if $(T - \lambda I)^k v = 0$ for some $k \ge 1$. The **generalized eigenspace** is $G(\lambda, T) = \bigcup_{k \ge 0} \operatorname{null}(T - \lambda I)^k$.

- $E(\lambda, T) \subseteq G(\lambda, T)$ (take $k = 1$). In the example, $E(0, N) = \operatorname{span}(e_1)$ but $G(0, N) = \mathbb{F}^2$.
- A union over all k is awkward. Props. 69–70 show it stops at a fixed power.
- An operator N is **nilpotent** if $N^k = 0$ for some k (Def. 39), e.g. N above. So $G(\lambda, T)$ is the part of V on which $T - \lambda I$ acts nilpotently.

$$v \xrightarrow{T-\lambda I} (T-\lambda I)v \xrightarrow{T-\lambda I} \cdots \xrightarrow{T-\lambda I} w \xrightarrow{T-\lambda I} 0$$

generalized eigenvector — last nonzero vector: an eigenvector

Repeatedly applying $T - \lambda I$ to a generalized eigenvector reaches 0. The last nonzero vector w has $(T - \lambda I)w = 0$, so it is an eigenvector (Prop. 71). Smallest case: $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $e_2 \mapsto e_1 \mapsto 0$.

Proposition (Prop. 69 / LADR 8.1, 8.2: the null-space chain)
$\{0\} = \operatorname{null} T^0 \subseteq \operatorname{null} T \subseteq \operatorname{null} T^2 \subseteq \cdots$. If $\operatorname{null} T^m = \operatorname{null} T^{m+1}$, then $\operatorname{null} T^m = \operatorname{null} T^{m+k}$ for all $k \ge 0$.

The idea

To show: (a) Each null space contains the one before. (b) Once two neighbours are equal, all later ones are equal.

(a) A vector killed by T^k stays killed by $T^{k+1} = T \cdot T^k$.

Only one equality, $\operatorname{null} T^m = \operatorname{null} T^{m+1}$. To use it on a vector $v \in \operatorname{null} T^{m+k+1}$, we need some vector in $\operatorname{null} T^{m+1}$ built from v.

Split the power: $T^{m+k+1}v = T^{m+1}(T^k v)$. So $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$, which says $T^{m+k}v = 0$. That is the equality one step higher; repeat.

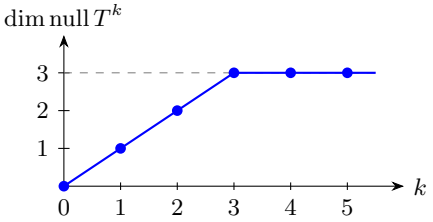
Example: The shift on $\mathbb{F}^3$ ($e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$) has $\dim \operatorname{null} T^k = 0, 1, 2, 3, 3, 3, \ldots$: it grows until two neighbours are equal, and after that all are equal.

Proof. 1. If $T^k v = 0$, then $T^{k+1}v = T(0) = 0$. So $\operatorname{null} T^k \subseteq \operatorname{null} T^{k+1}$.

2. Suppose $\operatorname{null} T^m = \operatorname{null} T^{m+1}$ and $T^{m+k+1}v = 0$. Then $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$.

3. So $T^{m+k}v = T^m(T^k v) = 0$. Hence $\operatorname{null} T^{m+k+1} \subseteq \operatorname{null} T^{m+k}$, and with step 1 they are equal. Induct on k.

□



$\operatorname{null} T \subsetneq \operatorname{null} T^2 \subsetneq \operatorname{null} T^3 = \operatorname{null} T^4 = \cdots$
each strict step adds at least 1 to the dimension,
which is at most $\dim V = 3$

For the shift on $\mathbb{F}^3$ the dimensions are $0, 1, 2, 3, 3, \ldots$. They can increase strictly at most $\dim V$ times, and once two neighbours are equal they stay equal (the chain lemma), so $\operatorname{null} T^n = \operatorname{null} T^{n+1}$ for $n = \dim V$.

Proposition (Prop. 70 / LADR 8.3, 8.20)
$\operatorname{null} T^n = \operatorname{null} T^{n+1}$. Hence $G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$, a T-invariant subspace.

The idea

To show: The chain has stopped by the power n.

Until it stops, each step raises the dimension by at least 1 (a strictly larger subspace has larger dimension). Dimensions are at most n. Once it stops, it stays stopped (Prop. 69).

Count: starting from dimension 0, $n+1$ strict steps would reach dimension $n+1 > n$. So one of the first $n+1$ steps is an equality, and from there on the chain is constant.

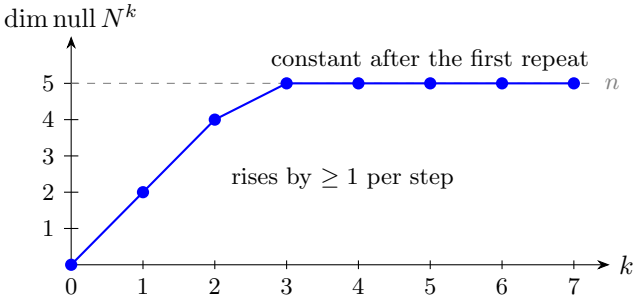
Why it matters: The union defining $G(\lambda, T)$ is then the single null space $\operatorname{null}(T - \lambda I)^n$. So it is a subspace, and it is invariant by (T5).

Proof. 1. Suppose $\operatorname{null} T^n \neq \operatorname{null} T^{n+1}$. By Prop. 69 no earlier pair is equal either.

2. Then
$$0 = \dim \operatorname{null} T^0 < \dim \operatorname{null} T^1 < \cdots < \dim \operatorname{null} T^{n+1} \leq n :$$
that is $n+2$ distinct integers in $\{0, \ldots, n\}$. Impossible.

3. So $\bigcup_k \operatorname{null}(T - \lambda I)^k = \operatorname{null}(T - \lambda I)^n$. It is invariant by (T5).

□



N on $\mathbb{F}^5$ with $Ne_1 = 0$, $Ne_2 = e_1$, $Ne_3 = e_2$, $Ne_4 = 0$, $Ne_5 = e_4$. Then $\operatorname{null} N = \operatorname{span}(e_1, e_4)$, $\operatorname{null} N^2 = \operatorname{span}(e_1, e_2, e_4, e_5)$, $\operatorname{null} N^3 = \mathbb{F}^5$, so $\dim \operatorname{null} N^k = 0, 2, 4, 5, 5, \ldots$ While the chain grows it gains at least 1 per step, and it cannot pass n, so it has stopped by $k = n$ (Prop. 70). Once two neighbours agree, all later ones agree (Prop. 69).

Proposition (LADR 8.4, 8.5: splitting off the nilpotent part)
$V = \operatorname{null} T^n \oplus \operatorname{range} T^n$, and T is injective on $\operatorname{range} T^n$.

The idea

To show: Split V into a part where T is nilpotent ($\operatorname{null} T^n$) and a part where T is invertible ($\operatorname{range} T^n$). Applied to $T - \lambda I$, this splits off $G(\lambda, T)$; that is the first step of Prop. 73.

$\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1). So by (T3) it is enough to show $\operatorname{null} T^n \cap \operatorname{range} T^n = \{0\}$.

The obvious first attempt: Power 1: is $\operatorname{null} T \cap \operatorname{range} T = \{0\}$? No. For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $e_1 = Ne_2$ is in the range and $Ne_1 = 0$. A nonzero vector of range N can lie in null N.

With the power n, $\operatorname{range} T^n$ meets $\operatorname{null} T^n$ only in 0, because the chain has stopped. If $v = T^n u$ and $T^n v = 0$, then $T^{2n} u = 0$, and $\operatorname{null} T^{2n} = \operatorname{null} T^n$ (Prop. 70) gives $T^n u = 0$, i.e. $v = 0$.

Proof. 1. Let $v = T^n u$ with $T^n v = 0$. Then $T^{2n} u = 0$, so $u \in \operatorname{null} T^{2n} = \operatorname{null} T^n$, so $v = 0$.

2. $\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1), so $V = \operatorname{null} T^n \oplus \operatorname{range} T^n$ (T3).

3. If $w \in \operatorname{range} T^n$ and $Tw = 0$, then $w \in \operatorname{null} T \subseteq \operatorname{null} T^n$, so $w = 0$.

□

Proposition (Prop. 71)
$G(\lambda, T) \neq \{0\} \iff \lambda$ is an eigenvalue of T.

The idea

To show: A nonzero generalized eigenvector for λ forces an eigenvector for λ.

Apply $T - \lambda I$ to v repeatedly: $v \mapsto (T - \lambda I)v \mapsto \cdots \mapsto 0$. The last nonzero vector in this list is sent to 0 by $T - \lambda I$, so it is an eigenvector.

Proof. If λ is an eigenvalue, $\{0\} \neq E(\lambda, T) \subseteq G(\lambda, T)$. Conversely, let $v \in G(\lambda, T)$, $v \neq 0$, and let $j \geq 0$ be the largest exponent with $w = (T - \lambda I)^j v \neq 0$ (it exists because $(T - \lambda I)^k v = 0$ for some k). Then $(T - \lambda I)w = 0$ and $w \neq 0$. □

Proposition (Prop. 72 / LADR 8.12)
Generalized eigenvectors $v_1, \ldots, v_m$ for distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ are independent.

The idea

To show: From $\sum a_i v_i = 0$, get each $a_k = 0$.

How independence of eigenvectors is proved (Prop. 54): apply an operator that kills every term except the k-th and is nonzero on the k-th. What is left is $a_k \cdot (\text{nonzero vector}) = 0$.

The obvious first attempt: $(T - \lambda_i I)^n$ kills v_i, so $S = \prod_{i \neq k} (T - \lambda_i I)^n$ kills every v_i with $i \neq k$. But it is not clear what S does to v_k, which is only a generalized eigenvector.

On an actual eigenvector w for λ_k, S is easy:

$$Sw = \prod_{i \neq k} (\lambda_k - \lambda_i)^n w \neq 0$$

(T7). So first turn v_k into an eigenvector by applying $(T - \lambda_k I)^N$, as in Prop. 71. The extra factor still gives $Sv_i = 0$ for $i \neq k$, because all the factors commute.

Proof. 1. Fix k; we show $a_k = 0$. Let N be the largest exponent with $w = (T - \lambda_k I)^N v_k \neq 0$ (it exists: $v_k \neq 0$, and some power of $T - \lambda_k I$ sends v_k to 0). By the choice of N, $(T - \lambda_k I)w = 0$, so w is an eigenvector: $Tw = \lambda_k w$.

2. On an eigenvector, each $T - \lambda_i I$ multiplies by the number $\lambda_k - \lambda_i$ (T7). So $(T - \lambda_i I)^n w = (\lambda_k - \lambda_i)^n w$, and this number is not 0 for $i \neq k$.

3. Let
$$S = \prod_{i \neq k} (T - \lambda_i I)^n \cdot (T - \lambda_k I)^N$$
and apply S to $\sum_i a_i v_i = 0$.
4. *Terms with $i \neq k$.* The factors of S commute (T5), so let $(T - \lambda_i I)^n$ act first on v_i. It gives 0, because $v_i \in G(\lambda_i, T) = \operatorname{null}(T - \lambda_i I)^n$ (Prop. 70). So $S v_i = 0$.
5. *The term with $i = k$.* $(T - \lambda_k I)^N v_k = w$, and then by step 2,
$$S v_k = \prod_{i \neq k} (\lambda_k - \lambda_i)^n \, w.$$
6. So
$$0 = a_k \prod_{i \neq k} (\lambda_k - \lambda_i)^n \, w.$$
The product is a nonzero number and $w \neq 0$, so $a_k = 0$.

□

Two consequences.

- $G(\lambda, T) \cap G(\mu, T) = \{0\}$ for $\lambda \neq \mu$: a nonzero v in both would make the list v, v independent (Prop. 72). So a generalized eigenvector belongs to only one eigenvalue.
- If U is invariant, $G(\lambda, T|_U) = G(\lambda, T) \cap U$, since $(T|_U - \lambda I)^k u = (T - \lambda I)^k u$ for $u \in U$.

Theorem (Prop. 73 / LADR 8.9, 8.22: generalized eigenspace decomposition)
Over $\mathbb{C}$, if $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T, then $V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$. In particular V has a basis of generalized eigenvectors of T.

The idea

To show: (a) The $G(\lambda_i, T)$ together span V. (b) The sum is direct.

(b) is Prop. 72: a relation $u_1 + \cdots + u_m = 0$ with $u_i \in G(\lambda_i, T)$ and some $u_i \neq 0$ would be a dependence among generalized eigenvectors for distinct eigenvalues.

The splitting result for $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$ with $U = \operatorname{range}(T - \lambda_1 I)^n$ invariant. U is smaller than V because $G(\lambda_1, T) \neq 0$.

Induct on $\dim V$. The theorem for $T|_U$ writes U as a sum of generalized eigenspaces of $T|_U$, and these lie inside those of T. The eigenvalue λ_1 does not come back in U, because $T - \lambda_1 I$ is injective on U.

Why $\mathbb{C}$. We need an eigenvalue λ_1 to start (T7). Over $\mathbb{R}$, the rotation $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has no eigenvalues, so all its generalized eigenspaces are 0.

Example. For
$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix},$$
$G(1, A) = \operatorname{span}(e_1, e_2)$ (Ex. 61 below) and $G(3, A) = E(3, A) = \operatorname{span}\big((3, 2, 4)\big)$. So $\mathbb{C}^3 = G(1, A) \oplus G(3, A)$.

Proof. 1. Induction on n; $n = 1$ is clear. Take an eigenvalue λ_1 (T7).

2. Splitting result for $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$, $U = \operatorname{range}(T - \lambda_1 I)^n$. U is invariant (T5), and $\dim U < n$ since $G(\lambda_1, T) \neq 0$ (Prop. 71).
3. *Which eigenvalues does $T|_U$ have?* An eigenvector of $T|_U$ is an eigenvector of T, so they are among $\lambda_1, \ldots, \lambda_m$. And λ_1 is not one of them: $T - \lambda_1 I$ is injective on $U = \operatorname{range}(T - \lambda_1 I)^n$ (part (b) of the splitting result, applied to $T - \lambda_1 I$).
4. By induction ($\dim U < n$), $U = \bigoplus_{i \geq 2} G(\lambda_i, T|_U)$. Each $G(\lambda_i, T|_U)$ is contained in $G(\lambda_i, T)$, because $(T|_U - \lambda_i I)^k u = (T - \lambda_i I)^k u$ for $u \in U$.

5. So the $G(\lambda_i, T)$ span V. Direct: if $u_1 + \cdots + u_m = 0$, $u_i \in G(\lambda_i, T)$, the nonzero u_i would be independent (Prop. 72); so all are 0.

□

On each $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with N_i nilpotent. So everything reduces to nilpotent operators (Ch. 15).

Definition (Def. 38)**. Algebraic multiplicity** of λ: $\dim G(\lambda, T)$. **Geometric multiplicity**: $\dim E(\lambda, T)$.

- geometric $\leq$ algebraic, since $E \subseteq G$.
- Over $\mathbb{C}$, the algebraic multiplicities add up to n (Prop. 73).
- Over $\mathbb{C}$, T is diagonalizable $\iff$ geometric = algebraic for every eigenvalue.

  Why. The sum $\sum E(\lambda_i, T)$ is direct (T7), and T is diagonalizable iff it is V, i.e. $\sum \dim E = n = \sum \dim G$. Since each $\dim E \leq \dim G$, the sums agree iff every term agrees.

Theorem (Prop. 74 / LADR 8.26, 8.29: Cayley–Hamilton)
Over $\mathbb{C}$, let $d_i = \dim G(\lambda_i, T)$ and $\chi_T(x) = \prod_i (x - \lambda_i)^{d_i}$ (the **characteristic polynomial**). Then $\chi_T(T) = 0$, so $m_T \mid \chi_T$. Also $\deg \chi_T = \sum d_i = n$ (Prop. 73), so $\deg m_T \leq n$.

The idea

To show: $\chi_T(T)v = 0$ for every v.

$V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73), so it is enough to take v in one $G(\lambda_i, T)$. There, $T - \lambda_i I$ is nilpotent.

Which power of $T - \lambda_i I$ kills $G(\lambda_i, T)$? Prop. 70, applied inside $G(\lambda_i, T)$, a space of dimension d_i, gives the power d_i. That is why the exponents in χ_T are the d_i.

The factors of $\chi_T(T)$ commute, so for $v \in G(\lambda_i, T)$ let $(T - \lambda_i I)^{d_i}$ act first.

Proof. 1. *Which power of $T - \lambda_i I$ is 0 on $G(\lambda_i, T)$?* Let $N = (T - \lambda_i I)|_{G(\lambda_i, T)}$, an operator on a space of dimension d_i. Every vector of that space is sent to 0 by some power of N. Prop. 70, applied to N on this d_i-dimensional space, says the power d_i is enough: $N^{d_i} = 0$.

2. For $v \in G(\lambda_i, T)$, the factors commute (T5), so the factor $(T - \lambda_i I)^{d_i}$ can act first, and it sends v to 0:

$$\chi_T(T)v = \prod_{j \neq i} (T - \lambda_j I)^{d_j} (T - \lambda_i I)^{d_i} v = 0.$$

3. The G's span V, so $\chi_T(T) = 0$; then $m_T \mid \chi_T$ (T11(a)) and $\deg \chi_T = \sum d_i = n$.

□

- For a matrix, $\chi_A(x) = \det(xI - A)$ (stated in the notes; needs determinants).
- 2×2: $\chi_A = x^2 - (\operatorname{tr} A)x + \det A$.

  Why. Multiplying out gives $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. If $A \neq cI$, m_A has degree 2 and divides both this polynomial and χ_A, so all three agree. If $A = cI$, both are $(x - c)^2$.
- χ_T and m_T have the same roots; the exponents in m_T can be smaller.

Proposition (Exponents of m_T over $\mathbb{C}$, M6)

Over $\mathbb{C}$, let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues and $d_i = \dim G(\lambda_i, T)$. Then $m_T = \prod_i (x - \lambda_i)^{s_i}$, where

$$s_i = \text{the least } k \text{ with } \operatorname{null}(T - \lambda_i I)^k = G(\lambda_i, T), \qquad 1 \le s_i \le d_i.$$

In particular $m_T \mid \chi_T$, and m_T has the same roots as χ_T.

The idea

To show: The exponent of each $x - \lambda_i$ in m_T.

V splits into the invariant pieces $G(\lambda_i, T)$ (Prop. 73), and the minimal polynomial of a direct sum is the lcm of the pieces' minimal polynomials (T11(e)).

So compute m on each piece separately. On $G(\lambda_i, T)$ the operator $T - \lambda_i I$ is nilpotent, so the minimal polynomial there is a power of $x - \lambda_i$, and the right power is the first one that is 0 on the piece.

Proof. 1. $(T - \lambda_i I)^n = 0$ on $G(\lambda_i, T)$ (Prop. 70), so $m_{T|_{G(\lambda_i,T)}}$ divides $(x - \lambda_i)^n$. Its roots are roots of $(x - \lambda_i)^n$, so it is $(x - \lambda_i)^{s_i}$ for some s_i (over $\mathbb{C}$ it splits into linear factors, T6).

2. s_i is the least k with $(T - \lambda_i I)^k = 0$ on $G(\lambda_i, T)$, i.e. with $\operatorname{null}(T - \lambda_i I)^k \supseteq G(\lambda_i, T)$. Since $\operatorname{null}(T - \lambda_i I)^k \subseteq G(\lambda_i, T)$ always, this means equality.

3. By Prop. 73 and T11(e), m_T is the least common multiple of the $(x - \lambda_i)^{s_i}$. The λ_i are distinct, so these factors have no common factor and the least common multiple is their product:

$$m_T = \operatorname{lcm}_i (x - \lambda_i)^{s_i} = \prod_i (x - \lambda_i)^{s_i}.$$

4. $s_i \ge 1$ since $G(\lambda_i, T) \ne 0$, and $s_i \le d_i$ by Prop. 70 applied inside the d_i-dimensional $G(\lambda_i, T)$.

□

Computing m_T from null-space dimensions. For each eigenvalue λ, compute

$$d_k = \dim \operatorname{null}(A - \lambda I)^k = n - \operatorname{rank}(A - \lambda I)^k$$

for $k = 1, 2, \dots$. The first k at which d_k stops growing (equivalently, reaches the algebraic multiplicity) is the exponent of $x - \lambda$.

Example

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

(Ex. 61). For $\lambda = 1$: $d_1 = 1$ and $d_2 = 2$, which is the number of 1s on the diagonal, so the exponent is 2. $\lambda = 3$ appears once, so its exponent is 1. So $m_A = (x - 1)^2 (x - 3)$.

- With the multiplicities d_i known, the candidate test for m_T from Week 7 needs only the candidates with $k_i \le d_i$, and χ_T itself never needs testing (Cayley–Hamilton). In the Method 2 example, 1 has multiplicity 1, so $(x - 3)(x - 1)^2$ was never a candidate, and $(x - 3)^2(x - 1) = \chi_A$ needed no check.
- For 3×3 A (using $\chi_A = \det(xI - A)$, stated in the notes; determinants are Ch. 25): $\chi_A = x^3 - (\operatorname{tr} A)x^2 + c_2 x - \det A$, where c_2 is the sum of the three 2×2 minors on the diagonal (rows and columns $\{1,2\}, \{1,3\}, \{2,3\}$). This gives the eigenvalues and multiplicities for Method 2 when A is not triangular.

Proposition (LADR 8.31: diagonal of a triangular matrix)
Over $\mathbb{C}$, if $\mathcal{M}(T)$ is upper-triangular, then $\dim G(\lambda, T)$ is the number of times λ appears on the diagonal.

The idea

To show: $\dim G(\lambda, T) = c_\lambda$, the number of times λ appears on the diagonal.

$$\dim G(\lambda, T) = \dim \operatorname{null}(T - \lambda I)^n = n - \operatorname{rank}(T - \lambda I)^n.$$

The matrix of $(T - \lambda I)^n$ is upper-triangular with diagonal entries $(a_{jj} - \lambda)^n$, nonzero in exactly $n - c_\lambda$ places.

The obvious first attempt: Compute the rank exactly. This is hard: columns with a 0 on the diagonal can still be nonzero and add to the rank.

Settle for an inequality. The $n - c_\lambda$ columns with nonzero diagonal entries are independent (T8), so rank $\geq n - c_\lambda$, i.e. $\dim G(\lambda, T) \leq c_\lambda$. Now add over all λ: both sides add up to n, so none of the inequalities can be strict.

Pattern: "Prove $\leq$ for each term, then compare totals" also proves the test above: T is diagonalizable if and only if each geometric multiplicity equals the algebraic multiplicity.

Proof. 1. Let c_λ be the count. $\mathcal{M}((T - \lambda I)^n)$ is triangular with diagonal $(a_{jj} - \lambda)^n$, nonzero in $n - c_\lambda$ places.

2. Those columns are independent (T8), so rank $\geq n - c_\lambda$ and $\dim G(\lambda, T) \leq c_\lambda$ (Prop. 70, T1).

3. By Prop. 73,
$$n = \sum_\lambda \dim G(\lambda, T) \leq \sum_\lambda c_\lambda = n,$$
so every inequality is an equality.

□

Example (Computing $G(\lambda, A)$ (Ex. 61))

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix},$$

$\lambda = 1$.

$\operatorname{null}(A - I) = \operatorname{span}(e_1)$, $\operatorname{null}(A - I)^2 = \operatorname{span}(e_1, e_2)$, $\operatorname{null}(A - I)^3 = \operatorname{span}(e_1, e_2)$.

2. Two consecutive null spaces agree, so the chain has stopped: $G(1, A) = \operatorname{span}(e_1, e_2)$.

3. Shortcut: 1 appears twice on the diagonal, so you can stop at dimension 2.

Shortcuts

- **Exponent 1 test.** The exponent of λ in m is 1 $\iff$ $\dim E(\lambda)$ equals the algebraic multiplicity
$$\iff \operatorname{rank}(A - \lambda I) = n - d_\lambda.$$
One rank computation per eigenvalue settles diagonalizability.
- **Never test χ itself** when computing m: it always works (Cayley–Hamilton). If every smaller candidate fails, $m = \chi$.
- **Smaller power.** $G(\lambda, T) = \operatorname{null}(T - \lambda I)^d$, d the algebraic multiplicity, and even $\operatorname{null}(T - \lambda I)^k$ with k the exponent of $x - \lambda$ in m_T (exponents of m_T, above). Why: $T - \lambda I$ is nilpotent on the d-dimensional space $G(\lambda, T)$, so Prop. 70 inside it gives the power d.
- **Know the target dimension first.** For a triangular matrix, $\dim G(\lambda)$ is the number of λ's on the diagonal (LADR 8.31); in general it is the multiplicity of λ as a root of $\chi_A = \det(xI - A)$. Stop the chain $\dim \operatorname{null}(A - \lambda I)^k$ as soon as it reaches that number.

- **Last one by subtraction.** The algebraic multiplicities add up to n, so the last one is n minus the others. And if the $\dim G(\lambda_i)$ you found already add up to n, there are no other eigenvalues.
- **Multiplicity 1.** If the algebraic multiplicity is 1, then $G(\lambda) = E(\lambda)$ (since $1 \le \dim E \le \dim G = 1$).
- **One eigenvalue.** Over $\mathbb{C}$, if λ is the only eigenvalue, $G(\lambda, T) = V$ and $(T - \lambda I)^n = 0$ with no computation.
- **Geometric multiplicity by rank.** $\dim E(\lambda) = n - \operatorname{rank}(A - \lambda I)$, and $\dim \operatorname{null}(A - \lambda I)^k = n - \operatorname{rank}(A - \lambda I)^k$.
- $G(\lambda)$ **as a range.** Over $\mathbb{C}$,
$$G(\lambda, T) = \operatorname{range} \prod_{\mu \neq \lambda} (T - \mu I)^{d_\mu},$$
and $\bigoplus_{\mu \neq \lambda} G(\mu, T)$ is the null space of that product. Useful when the other eigenvalues are simple: for the A of Ex. 61, $G(1, A) = \operatorname{range}(A - 3I) = \operatorname{span}(e_1, e_2)$. Why, with P the product and $H = \bigoplus_{\mu \neq \lambda} G(\mu, T)$:
  1. $P = 0$ on each $G(\mu, T)$, $\mu \neq \lambda$: let $(T - \mu I)^{d_\mu}$ act first (Prop. 74, step 1).
  2. P is injective on $G(\lambda, T)$: each $T - \mu I$ is, since a nonzero $v \in G(\lambda, T)$ with $Tv = \mu v$ would lie in $G(\lambda, T) \cap G(\mu, T) = 0$.
  3. $G(\lambda, T)$ is invariant (T5), so P maps it injectively into itself, hence onto itself (T2).
  4. $V = G(\lambda, T) \oplus H$ (Prop. 73) and $P = 0$ on H, so $\operatorname{range} P = G(\lambda, T)$ and $\operatorname{null} P = H$.
- **Commuting operators.** If $ST = TS$, then S maps $G(\lambda, T)$ into itself (G2): $(T - \lambda I)^n Sv = S(T - \lambda I)^n v = 0$.

Traps

- $G(\lambda, T)$ can be strictly bigger than $E(\lambda, T)$. Over $\mathbb{C}$ they agree for all λ exactly when T is diagonalizable.
- The exponents in m_T are at most the algebraic multiplicities, and they are not the geometric multiplicities: on $\mathbb{F}^4$ with $Ne_2 = e_1$ and $Ne_1 = Ne_3 = Ne_4 = 0$, $m_N = x^2$ while $\dim E(0) = 3$ and $\dim G(0) = 4$.
- A sum of generalized eigenvectors for different eigenvalues is not a generalized eigenvector (Prop. 72 makes the pieces independent).
- Prop. 73 needs eigenvalues: true over $\mathbb{C}$; over $\mathbb{R}$ only when m_T splits. (If m_T splits, every $T|_U$ with $U \neq 0$ invariant has an eigenvalue, because $m_{T|_U} \mid m_T$ (T11(d)) and its roots are eigenvalues (T11(b)); so the proof of Prop. 73 still works.)
- $V = \operatorname{null} T \oplus \operatorname{range} T$ can fail: for $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\operatorname{null} N = \operatorname{range} N = \operatorname{span}(e_1)$. The splitting needs the power n.
- The power n always suffices; the chain often stops sooner.

§2.2 Nilpotent operators and Jordan form (Ch. 15)

The question. A complex operator is a direct sum of pieces $\lambda I + N$ with N nilpotent (Prop. 73), and λI looks the same in every basis. So we only need a good basis for a nilpotent N.

- Simplest nilpotent: a shift $v \mapsto Nv \mapsto \cdots \mapsto N^d v \mapsto 0$. In the basis $N^d v, \ldots, Nv, v$ its matrix has 1s just above the diagonal.
- The hope: every nilpotent operator is a direct sum of shifts.
- A basis of generalized eigenvectors is not enough: e_1, e_2, e_3 are generalized eigenvectors for 1 of I,
$$B = \begin{pmatrix} 3 & -1 & -1 \\ 2 & 0 & -1 \\ 2 & -1 & 0 \end{pmatrix} \quad \text{and} \quad C = \begin{pmatrix} 3 & -2 & 0 \\ 1 & 1 & -1 \\ 0 & 2 & -1 \end{pmatrix},$$
yet $\dim E(1) = 3, 2, 1$, so none are similar.

Definition (Def. 39)**.** N is **nilpotent** if $N^k = 0$ for some k; the least such k is the **nilpotency index**.

- $N^n = 0$: $V = G(0, N) = \operatorname{null} N^n$ (Prop. 70). So $m_N = x^k$, $k \le n$.
- The only eigenvalue is 0 (if $V \neq 0$): $Nv = \lambda v \Rightarrow 0 = N^k v = \lambda^k v$. Over $\mathbb{C}$, conversely, only eigenvalue $0 \Rightarrow V = G(0, N)$ (Prop. 73).
- $N \neq 0$ is not diagonalizable: a diagonal matrix with all eigenvalues 0 is 0.
- Index $= \deg m_N \le n$. HW8 P1: D on $\mathcal{P}_n(\mathbb{C})$ has index $n+1$, since $D^n x^n = n! \neq 0$ and $D^{n+1} = 0$.
- $(I - N)^{-1} = I + N + \cdots + N^{n-1}$: the product telescopes to $I - N^n = I$.

Proposition (LADR 8.18: nilpotent operators are strictly upper-triangular)
If N is nilpotent, then V has a basis in which $\mathcal{M}(N)$ is upper-triangular with every diagonal entry 0.

The idea

To show: A basis $v_1, \ldots, v_n$ with $Nv_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$ for every j. That is exactly an upper-triangular matrix with zero diagonal.

If $N^k v = 0$, then $N^{k-1}(Nv) = 0$. So N maps $\operatorname{null} N^k$ into $\operatorname{null} N^{k-1}$.

List vectors by the least power of N that sends them to 0: a basis of $\operatorname{null} N$, extended to a basis of $\operatorname{null} N^2$, and so on up to $\operatorname{null} N^n = V$. A vector added at stage k is sent into $\operatorname{null} N^{k-1}$, which is spanned by vectors listed earlier.

Example: $N = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ ($Ne_1 = e_2$, $Ne_2 = 0$): $\operatorname{null} N = \operatorname{span}(e_2)$, so list e_2 first. In the basis e_2, e_1 the matrix is $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$.

Proof. 1. Choose a basis of $\operatorname{null} N$, extend it to a basis of $\operatorname{null} N^2$, and so on up to $\operatorname{null} N^n = V$ (Props. 69, 70).

2. If v was added for $\operatorname{null} N^k$, then $N^{k-1}(Nv) = N^k v = 0$, so $Nv \in \operatorname{null} N^{k-1}$, the span of the earlier layers (for $k = 1$, $Nv = 0$).

3. So each basis vector is sent into the span of the vectors listed before it: column j has nonzero entries only above the diagonal.

□

Theorem (LADR 8.37: block diagonal form)
Over $\mathbb{C}$, let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T and $d_i = \dim G(\lambda_i, T)$. There is a basis in which

$$\mathcal{M}(T) = \begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_m \end{pmatrix},$$

where each A_i is a $d_i \times d_i$ upper-triangular matrix with every diagonal entry equal to λ_i.

The idea

To show: A block diagonal matrix whose i-th block is upper-triangular with every diagonal entry λ_i.

Prop. 73 splits V into invariant pieces $G(\lambda_i, T)$, and invariant pieces give a block diagonal matrix (T9). So only the basis inside each piece is left to choose.

On $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with N_i nilpotent, and $\lambda_i I$ has the matrix $\lambda_i I$ in every basis. So choose the basis that makes N_i strictly upper-triangular (LADR 8.18); adding $\lambda_i I$ puts λ_i on the diagonal.

Proof. 1. $N_i = (T - \lambda_i I)|_{G(\lambda_i, T)}$ is nilpotent, since $G(\lambda_i, T) = \operatorname{null}(T - \lambda_i I)^n$ (Prop. 70). By LADR 8.18, $G(\lambda_i, T)$ has a basis in which $\mathcal{M}(N_i)$ is upper-triangular with zero diagonal.

2. In that basis $\mathcal{M}(T|_{G(\lambda_i, T)}) = \lambda_i I + \mathcal{M}(N_i)$: upper-triangular, diagonal entries λ_i, size d_i.

3. $V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73) with each piece invariant, so concatenating the bases gives a block diagonal matrix (T9).

□

$$\mathcal{M}(T) = \left(\begin{array}{ccc|cc} \lambda_1 & * & * & 0 & 0 \\ 0 & \lambda_1 & * & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 \\ \hline 0 & 0 & 0 & \lambda_2 & * \\ 0 & 0 & 0 & 0 & \lambda_2 \end{array}\right)$$

columns 1–3: a basis of $G(\lambda_1, T)$, $d_1 = 3$
columns 4–5: a basis of $G(\lambda_2, T)$, $d_2 = 2$

The zero blocks come from splitting V into invariant pieces (Prop. 73, T9). Each diagonal block is $\lambda_i I$ plus a strictly upper-triangular nilpotent part (LADR 8.18).

Proposition (Chain lemma)
If $N^d v \neq 0$ and $N^{d+1} v = 0$, then $v, Nv, \ldots, N^d v$ are independent.

The idea

To show: From $\sum_{j=0}^{d} c_j N^j v = 0$, get every $c_j = 0$.

Applying N moves every term one step down the chain, and a term that is moved past $N^d v$ becomes 0.

Apply N^d: every term except $c_0 v$ is moved past $N^d v$ and becomes 0, leaving $c_0 N^d v = 0$, so $c_0 = 0$. Then N^{d-1} leaves only the c_1 term, and so on. Same idea as Prop. 72: apply a map that sends every term but one to 0.

Proof. Apply N^d to $\sum_j c_j N^j v = 0$: get $c_0 N^d v = 0$, so $c_0 = 0$. Then apply N^{d-1} to get $c_1 = 0$, and so on. □

$$v \xrightarrow{N} Nv \xrightarrow{N} N^2 v \neq 0 \xrightarrow{N} 0$$

coefficient c_0 coefficient c_1 coefficient c_2

The case $d = 2$. Applying N^2 to $c_0 v + c_1 Nv + c_2 N^2 v$ moves each term two steps right. Only the c_0 term lands on $N^2 v \neq 0$; the other two land on 0. So $c_0 N^2 v = 0$ forces $c_0 = 0$.

Theorem (Prop. 75 / LADR 8.45: nilpotent normal form)
A nilpotent N has a basis made of chains $u, Nu, \ldots, N^k u$ ($N^{k+1} u = 0$). Listing each chain as $N^k u, \ldots, Nu, u$, the matrix $\mathcal{M}(N)$ has only 1s and 0s directly above the diagonal and 0 everywhere else.

The idea

The smallest case that shows everything. On $\mathbb{F}^3$ let $Ne_3 = e_2$, $Ne_2 = 0$, $Ne_1 = 0$. A chain is a list $u, Nu, N^2 u, \ldots$ stopped at its last nonzero vector. Here e_3, e_2 is a chain of length 2 and e_1 is a chain of length 1, and together they are a basis. In the order e_2, e_3, e_1 the matrix of N has a single 1 (from $Ne_3 = e_2$) and zeros elsewhere. The theorem says every nilpotent operator has a basis like this.

Plan of the proof. *Where to look for the chains.* Induction needs a smaller space that N maps into itself. $U = \operatorname{range} N$ works: it is invariant, and it is smaller than V because N is not injective (T2). In the example $U = \operatorname{span}(e_2)$, with chain basis e_2.

What a chain basis of U is missing. Compare it with what V needs.

(1) Every chain of U starts at a vector $v_i \in \operatorname{range} N$, so $v_i = Nu_i$ for some u_i. That u_i is not in the list yet. Put it on top; the chain becomes one longer. (Example: $e_2 = Ne_3$, so add e_3.)

(2) A vector that N sends to 0 but that is not in $\operatorname{range} N$ lies on no chain. (Example: e_1.) Such vectors must be added as chains of length 1.

Which vectors of $\operatorname{null} N$ *are already in the list?* Only the bottoms of the chains: the bottom is the one vector of a chain that N sends to 0. So extend the bottoms to a basis of $\operatorname{null} N$ by new vectors $w_1, \ldots, w_l$.

What is left to check. That the new list has n vectors, and that it is independent. The count is rank–nullity: the old list has $\dim \operatorname{range} N$ vectors, and we added one top per chain (r of them) and l vectors w, where $r + l = \dim \operatorname{null} N$. Independence is the step that needs care. The plan: apply N to a relation. That removes the w's and the bottoms and turns everything else into vectors of the old chain basis of U, which we know is independent.

Proof. 1. If $N = 0$, any basis works (chains of length 1). Otherwise $U = \operatorname{range} N$ is invariant with $0 < \dim U < n$. By induction U has a chain basis $v_i, Nv_i, \dots, N^{d_i} v_i$ for $i = 1, \dots, r$, with $N^{d_i+1} v_i = 0$. Write $b_i = N^{d_i} v_i$ for the bottoms.

2. *The bottoms are a basis of* $\operatorname{null} N \cap U$. They lie in $\operatorname{null} N$ and in U, and they are independent (part of a basis). They span: take $x \in U$ with $Nx = 0$ and write $x = \sum_{i,j} c_{ij} N^j v_i$. Then $0 = Nx = \sum_{j<d_i} c_{ij} N^{j+1} v_i$ (the terms with $j = d_i$ are 0). The vectors $N^{j+1} v_i$ with $j < d_i$ are different members of the chain basis, so $c_{ij} = 0$ for $j < d_i$, and $x = \sum_i c_{id_i} b_i$.

3. Choose u_i with $Nu_i = v_i$ (possible since $v_i \in \operatorname{range} N$). Extend $b_1, \dots, b_r$ by $w_1, \dots, w_l$ to a basis of $\operatorname{null} N$. So $\dim \operatorname{null} N = r + l$.

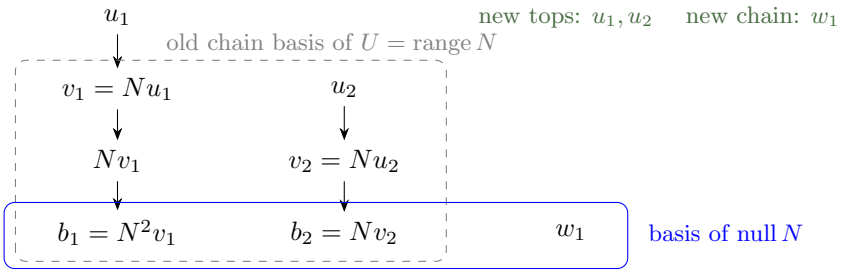
4. *The new list* is $u_i, v_i, Nv_i, \dots, b_i$ for each i, together with $w_1, \dots, w_l$. *Count,* using rank–nullity (T1) at the last step:
$$\sum_i (d_i + 2) + l = \sum_i (d_i + 1) + r + l = \dim \operatorname{range} N + \dim \operatorname{null} N = n.$$

5. *Independence, first half.* Suppose $\sum_i a_i u_i + \sum_{i,j} c_{ij} N^j v_i + \sum_k e_k w_k = 0$. Apply N. We have $Nu_i = v_i$; $N(N^j v_i) = N^{j+1} v_i$, which is 0 for $j = d_i$; and $Nw_k = 0$. So
$$\sum_i a_i v_i + \sum_{j<d_i} c_{ij} N^{j+1} v_i = 0.$$
The vectors here are different members of the chain basis of U (the tops v_i, and the vectors below them). So every $a_i = 0$, and $c_{ij} = 0$ for $j < d_i$.

6. *Independence, second half.* What is left of the relation is $\sum_i c_{id_i} b_i + \sum_k e_k w_k = 0$. The b_i and w_k together are a basis of $\operatorname{null} N$ (step 3), so these coefficients are 0 too. This is the reason the w's were chosen to extend the bottoms: a w that was a combination of chain vectors would, by step 5, be a combination of the bottoms, and then bottoms and w's would be dependent.

7. So the new list is n independent vectors, a basis (T3), and it consists of chains. Listing a chain as $N^k u, \dots, Nu, u$, N sends each vector to the one before it and the first to 0. That is a column with a single 1 directly above the diagonal, or a zero column.

□



Two chains of U (inside the dashed box) with $d_1 = 2$, $d_2 = 1$. Each gets a new top u_i with $Nu_i = v_i$. The bottoms b_1, b_2 are extended by w_1 to a basis of $\operatorname{null} N$ (bottom row). Arrows are N; the bottom row is sent to 0.

Definition (Def. 40)**. Jordan block**: $J_k(\lambda) = \lambda I + N_k$ ($k \times k$, 1s on the superdiagonal), so $Je_1 = \lambda e_1$, $Je_j = \lambda e_j + e_{j-1}$. **Jordan normal form**: block diagonal with Jordan blocks. **Jordan basis**: a basis giving Jordan form.

Example (HW8 P1). D on $\mathcal{P}_n(\mathbb{C})$ is the single block $J_{n+1}(0)$ in the basis $1, x, x^2/2!, \dots, x^n/n!$, because D sends each basis vector to the previous one:
$$D\frac{x^k}{k!} = \frac{x^{k-1}}{(k-1)!}.$$

Theorem (Prop. 76 / LADR 8.46: Jordan normal form)

Every operator on a finite-dimensional complex space has a Jordan basis.

The idea

To show: A basis in which T is block diagonal with Jordan blocks.

Prop. 73 splits V into invariant pieces $G(\lambda_i, T)$, so the matrix is block diagonal (T9). On each piece $T = \lambda_i I + N_i$ with N_i nilpotent, and $\lambda_i I$ has the same matrix in every basis.

Use the chain basis of N_i (Prop. 75) in each piece: N_i becomes a sum of blocks N_k, so T becomes a sum of blocks $\lambda_i I + N_k = J_k(\lambda_i)$. This is the proof of LADR 8.37 with the chain basis in place of the triangular one.

Proof. On each invariant $G(\lambda_i, T)$, $N_i = (T - \lambda_i I)|_G$ is nilpotent. Prop. 75 gives a basis where N_i is a sum of blocks N_k, so $T = \lambda_i I + N_i$ is a sum of blocks $J_k(\lambda_i)$. Concatenate the bases (T9). □

Compared with LADR 8.37. Both start by splitting V into the $G(\lambda_i, T)$. LADR 8.37 only makes each block upper-triangular with λ_i on the diagonal; Jordan form also makes every entry above the diagonal 0, except the entries directly above it, which are 0 or 1. That needs the chain construction of Prop. 75.

Reading the Jordan form. For each eigenvalue λ:

- number of blocks for λ = geometric multiplicity $\dim E(\lambda, T)$;
- total size of those blocks = algebraic multiplicity $\dim G(\lambda, T)$;
- largest block for λ = exponent of $(x - \lambda)$ in m_T;
- $\chi_T = \prod (x - \lambda)^{\text{alg. mult.}}$, $m_T = \prod (x - \lambda)^{\text{largest block}}$;
- diagonalizable $\iff$ all blocks 1×1 $\iff$ m_T has no repeated root.

Why.

1. One block $J_k(\lambda)$: $J_k(\lambda) - \lambda I = N_k$ moves each basis vector one place ($e_j \mapsto e_{j-1}$, $e_1 \mapsto 0$), so N_k^j moves it j places and $\dim \operatorname{null} N_k^j = \min(j, k)$. Hence $E = \operatorname{span}(e_1)$ (one block, dimension 1), G is the whole block (size k), and $m = \chi = (x - \lambda)^k$.
2. A block $J_j(\mu)$ with $\mu \neq \lambda$ adds nothing to $\operatorname{null}(T - \lambda I)^k$: $(J_j(\mu) - \lambda I) = (\mu - \lambda) I + N_j$ is invertible (a nilpotent has only the eigenvalue 0, so $N_j v = (\lambda - \mu) v$ forces $v = 0$), and so are its powers.
3. For the whole matrix, null spaces add over the blocks (T9) and m is the lcm of the blocks' minimal polynomials (T11(e)). Adding up gives each line of the box.

Theorem (Prop. 77: counting blocks)

Over $\mathbb{C}$, let $d_k = \dim \operatorname{null}(T - \lambda I)^k$ ($d_0 = 0$). For $k \geq 1$:

- number of blocks for λ of size $\geq k = d_k - d_{k-1}$;
- number of size exactly $k = 2d_k - d_{k-1} - d_{k+1}$.

So the Jordan form is unique up to order, and two complex matrices are similar iff they have the same Jordan form.

The idea

The numbers d_k are defined from T alone, without choosing a basis; so it is enough to express the block sizes through them. Compute d_k in a Jordan basis. Null spaces split block by block (T9), so d_k is a sum over the blocks. For one block, $J_j(\lambda) - \lambda I$ is a shift, and the null space of its k-th power has dimension $\min(k, j)$: it rises by 1 per step until $k = j$, then stays. So from $k - 1$ to k, exactly the blocks of size $\geq k$ add 1 (blocks for other eigenvalues add 0).

Example. For the blocks $J_3(0), J_1(0)$, $d_1, d_2, d_3 = 2, 3, 4$. So there are 2 blocks of size ≥ 1, 1 of size ≥ 2, and 1 of size ≥ 3.

Proof. 1. In a Jordan basis, $d_k = \sum_{\text{blocks}} \dim \operatorname{null}(J - \lambda I)^k$. Blocks for $\mu \neq \lambda$ contribute 0; $J_j(\lambda)$ contributes $\min(k, j)$.

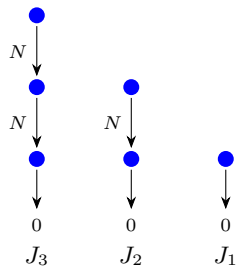
2. $\min(k, j) - \min(k-1, j)$ is 1 if $j \geq k$, else 0. Summing:

$$d_k - d_{k-1} = \#\{\text{blocks with } j \geq k\}.$$

Exactly k: $(d_k - d_{k-1}) - (d_{k+1} - d_k)$.

3. Similar matrices are one operator in two bases (T10), so they have the same d_k and the same Jordan form. Conversely, matrices with the same Jordan form are both similar to it.

□



row 3: 1 dot, $d_3 = 5 + 1 = 6$

row 2: 2 dots, $d_2 = 3 + 2 = 5$

row 1: 3 dots, $d_1 = 3$

A nilpotent N with blocks $J_3(0), J_2(0), J_1(0)$. Each column is one chain, read downward; the bottom dot is an eigenvector. N^k sends exactly the dots in the bottom k rows to 0, so d_k is the number of dots in rows $1, \ldots, k$, and $d_k - d_{k-1}$ is the number of dots in row k, i.e. the number of blocks of size $\geq k$.

Example (Jordan form from null-space dimensions (Ex. 63))

$T \in \mathcal{L}(\mathbb{C}^8)$ with eigenvalues $1, 2$. Method: list $d_0, d_1, \ldots$, take differences, then differences of differences.

λ	d_0	d_1	d_2	d_3	d_4	blocks
1	0	2	4	4		sizes ≥ 1: 2, ≥ 2: 2, ≥ 3: 0 $\Rightarrow J_2(1), J_2(1)$
2	0	2	3	4	4	sizes ≥ 1: 2, ≥ 2: 1, ≥ 3: 1, ≥ 4: 0 $\Rightarrow J_3(2), J_1(2)$

($d_4 = 4$ for $\lambda = 2$ because $\dim G(2, T) = 8 - 4$.) Answer: $J_2(1) \oplus J_2(1) \oplus J_3(2) \oplus J_1(2)$.

Example (Finding a Jordan basis (Ex. 62))

The matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

has only the eigenvalue 1, and $d_1, d_2, d_3 = 1, 2, 3$, so it has one block $J_3(1)$.

1. Pick the top: $v_3 \in \operatorname{null}(A - I)^3$, $v_3 \notin \operatorname{null}(A - I)^2$. Here $v_3 = e_3$.
2. Go down: $v_2 = (A - I)v_3 = (3, 2, 0)$, $v_1 = (A - I)v_2 = (4, 0, 0)$ (an eigenvector).
3. Check: $Av_1 = v_1$, $Av_2 = v_2 + v_1$, $Av_3 = v_3 + v_2$. That is $J_3(1)$.

$$v_3 = (0, 0, 1) \xrightarrow{A - I} v_2 = (3, 2, 0) \xrightarrow{A - I} v_1 = (4, 0, 0) \xrightarrow{A - I} 0$$

top: not in $\operatorname{null}(A - I)^2$ — eigenvector

Start at the top and apply $A - I$ until the next step gives 0. In the basis (v_1, v_2, v_3), read from the bottom of the chain, each vector is sent to the one before it, which puts the 1s just above the diagonal.

Why start at the top: the top vector determines the whole chain, and a top outside $\operatorname{null}(A - \lambda I)^{k-1}$ guarantees full length k (chain lemma).

Several blocks: get the block sizes first (Prop. 77). For each block of size j, largest first, pick a top $v \in \operatorname{null}(A-\lambda I)^j \setminus \operatorname{null}(A-\lambda I)^{j-1}$ and keep its chain only if its bottom $(A-\lambda I)^{j-1}v$ is independent of the bottoms already chosen. (Independent of the vectors so far is not enough: in $J_3(0)\oplus J_2(0)$ with chains e_1, e_2, e_3 and f_1, f_2, the top f_1+e_2 gives a chain ending at e_1 again.)

Powers. $J = \lambda I + N$ with λI and N commuting, so the binomial theorem applies and $N^j = 0$ for $j \geq k$:

$$J_k(\lambda)^m = \sum_{j=0}^{\min(m,k-1)} \binom{m}{j} \lambda^{m-j} N^j, \qquad J_3(\lambda)^m = \begin{pmatrix} \lambda^m & m\lambda^{m-1} & \binom{m}{2}\lambda^{m-2} \\ 0 & \lambda^m & m\lambda^{m-1} \\ 0 & 0 & \lambda^m \end{pmatrix}.$$

- Example: $J_3(1)^{100}$ has 100 and 4950 above the diagonal.
- Any Jordan matrix is $J = D + N$, D diagonal, N nilpotent, $DN = ND$ (on each block $D = \lambda I$); this is the Jordan–Chevalley decomposition. If J is invertible, also $J = DU$ with $U = I + D^{-1}N$, $DU = UD$.
- **Square roots (LADR 8.39–8.41).** $I+N$ has a square root: choose a_j with $(I+a_1N+\cdots+a_{n-1}N^{n-1})^2 = I+N$ by matching coefficients ($2a_1 = 1$, $2a_2 + a_1^2 = 0$, …); powers N^n and up vanish. This is the series

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots$$

  cut off. Over $\mathbb{C}$, every invertible operator has a square root (apply this to $\lambda_i(I + N_i/\lambda_i)$ on each $G(\lambda_i, T)$).

Traps

- χ_T and m_T do not fix the Jordan form: $J_2(0)\oplus J_2(0)$ and $J_2(0)\oplus J_1(0)\oplus J_1(0)$ both have $\chi = x^4$, $m = x^2$.
- Adding the number of blocks is still not enough: $J_3 \oplus J_3 \oplus J_1$ and $J_3 \oplus J_2 \oplus J_2$ (all for 0) agree in χ, m and block count, but $d_2 = 5$ vs. 6. You need all the d_k.
- Order each chain from the bottom (the eigenvector) up; reversed, the 1s go below the diagonal.
- Jordan form needs $\mathbb{C}$ (or split m_T): the rotation of $\mathbb{R}^2$ has none over $\mathbb{R}$.
- The exponent of λ in m_T is the size of the largest block for λ, not the number of blocks (that is $\dim E(\lambda, T)$).

§3 Problem set

HW8 P1. Let $D\colon \mathcal{P}_n(\mathbb{C}) \to \mathcal{P}_n(\mathbb{C})$ be differentiation.

(a) Prove that D is nilpotent and determine its nilpotency index.

(b) Determine the geometric and algebraic multiplicities of the eigenvalue 0.

(c) Find a basis with respect to which the matrix of D is a single Jordan block.

HW8 P2. Construct explicit matrices having the following spectral data.

(a) Eigenvalue 2 has algebraic multiplicity 3 and geometric multiplicity 1; eigenvalue -1 has algebraic multiplicity 2 and geometric multiplicity 2.

(b) The same algebraic multiplicities as in (a), but the geometric multiplicity of 2 is 2.

For each matrix, give its minimal polynomial.

HW8 P3. Let

$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

(a) Compute $\operatorname{null}(J-I)$, $\operatorname{null}((J-I)^2)$, and $\operatorname{null}((J-I)^3)$.

(b) Determine the generalized eigenspace $G(1, J)$ and the geometric and algebraic multiplicities of 1.

(c) Compute J^m for every positive integer m.

HW8 P4. Let N be nilpotent with Jordan form consisting of one block of size 3 and one block of size 2.

(a) Compute $\dim \operatorname{null} N^k$ for $k = 0, 1, 2, 3, 4$.

(b) Suppose instead that you are only told $\dim \operatorname{null} N = 2$, $\dim \operatorname{null} N^2 = 4$, $\dim \operatorname{null} N^3 = 5$. Explain how these numbers force the same two block sizes.

HW8 P5. (optional) Let $T \in \mathcal{L}(\mathbb{C}^6)$ have eigenvalues 1 and 2, and suppose

$$\dim \operatorname{null}(T - I) = 1, \quad \dim \operatorname{null}((T - I)^2) = 2, \quad \dim \operatorname{null}((T - I)^3) = 2,$$

while

$$\dim \operatorname{null}(T - 2I) = 2, \quad \dim \operatorname{null}((T - 2I)^2) = 3, \quad \dim \operatorname{null}((T - 2I)^3) = 4, \quad \dim \operatorname{null}((T - 2I)^4) = 4.$$

Determine the Jordan normal form of T, up to reordering of the blocks.

HW8 P6. (optional) Suppose an operator has Jordan blocks $J_3(4)$, $J_2(4)$, $J_1(-1)$. Determine:

(a) its characteristic polynomial;

(b) its minimal polynomial;

(c) the geometric and algebraic multiplicities of each eigenvalue;

(d) whether it is diagonalizable.

HW8 P7. (optional) Prove that for a nilpotent operator N, $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$ is the number of Jordan blocks of size at least k.

G1. Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

act on $\mathbb{C}^3$. Find the eigenvalues, $E(2, A)$, $G(2, A)$, the Jordan form, a Jordan basis, and m_A.

G2. Let V be finite-dimensional and $S, T \in \mathcal{L}(V)$ with $ST = TS$. Prove that $G(\lambda, T)$ is invariant under S.

G3.

(a) Let N be nilpotent on a finite-dimensional $V \neq \{0\}$. Prove that $\lambda I + N$ is invertible if and only if $\lambda \neq 0$.

(b) Let V be a finite-dimensional complex vector space and suppose $T \in \mathcal{L}(V)$ has only one eigenvalue λ. Prove that $T - \lambda I$ is nilpotent.

G4. Find all possible Jordan forms of $T \in \mathcal{L}(\mathbb{C}^4)$ with $\chi_T = (x - 2)^4$ and $m_T = (x - 2)^2$. Which single number tells them apart?

G5. Let V be a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that if every generalized eigenvector of T is an eigenvector, then T is diagonalizable.